

1

LOW CONTRAST IMAGE

Scenario: A grayscale image appears dull with very little difference between object and background intensities.

(a) Interpretation & Cause

Low contrast occurs when the range of gray levels used in the image is narrow. Pixel intensities are clustered in a small range, making details difficult to distinguish.

Low Contrast
(Input)



(b) Key Issues

- Narrow dynamic range
- Poor visibility of details
- Objects blend with background
- Loss of important information

Enhanced
(Output)

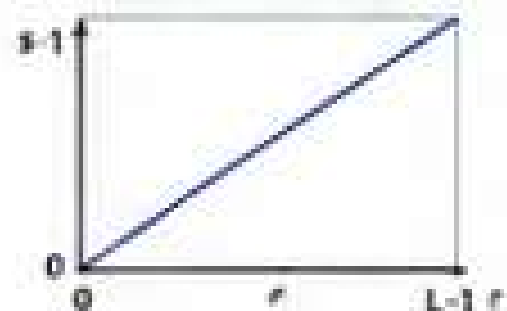


(c) Applied Technique: Contrast Stretching (Linear)

Formula:

$$s = \frac{r - r_{\min}}{r_{\max} - r_{\min}} (L - 1)$$

Transformation Curve



(d) Justification

Contrast stretching uses the full dynamic range (0 to L-1), expanding pixel values and improving visibility of both dark and bright regions.

(e) Algorithm / Steps

1. Find $r_{\min}$ and $r_{\max}$ of the image
2. Apply linear transformation
3. Map values to full range [0, L-1]

Observation: Details in shadows and highlights become visible; overall contrast is improved.

Conclusion:

Contrast stretching for low contrast images.



2

UNEVEN ILLUMINATION

Scenario: An image captured under non-uniform lighting shows bright and dark patches.

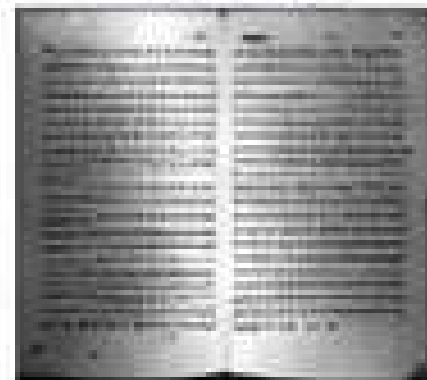
(a) Interpretation & Impact

Uneven illumination causes certain regions to appear too bright or too dark, reducing the visibility of features and making analysis inaccurate.

(b) Key Factors

- Varying light source intensity
- Shadows and reflections
- Non-uniform sensor response
- Background variations

Input Image



(c) Applied Technique: Background Correction using Homomorphic Filtering

Idea: Separate illumination (low frequency) from reflectance (high frequency) and normalize illumination.

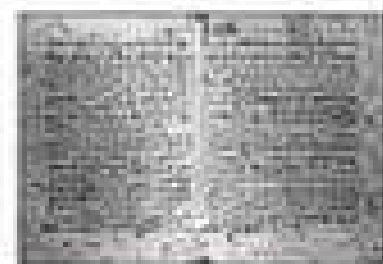


(d) Justification

Homomorphic filtering works in frequency domain to suppress low-frequency illumination variations and preserves high-frequency details.

(e) Result

Corrected Image



Observation: Illumination is more uniform and details are clearly visible across the image.



3

SALT-AND-PEPPER NOISE

Scenario: A scanned document contains salt-and-pepper noise affecting readability.

(a) Interpretation & Noise Characteristics

Salt-and-pepper noise appears as random white (salt) and black (pepper) pixels. It is an impulse noise caused by bit errors during acquisition or transmission.

(b) Key Issues

- Random black and white pixels
- Obscures text and fine details
- Reduces readability
- Difficult for thresholding and segmentation

Noisy Image (Input)

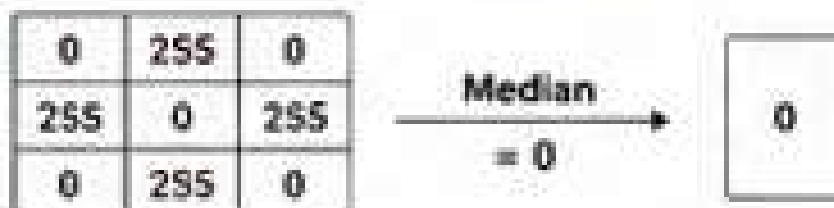
Original document with significant salt-and-pepper noise. Text is mostly obscured by random black and white pixels, making it difficult to read.

Filtered Image (Output)

Document after applying a median filter. The noise has been significantly reduced, and the text is now clearly visible.

(c) Applied Technique: Median Filtering (3x3)

Process: Median filter replaces the center pixel by the median of neighborhood values.



(d) Justification

Median filter is highly effective for impulse noise while preserving edges and fine details better than mean filter.

(e) Algorithm / Steps

1. Select 3x3 (or 5x5) window
2. Sort pixel values
3. Replace center with median

Observation: Salt-and-pepper noise is while text and edges are preserved.

Conclusion:

Median filtering is best for impulse noise.



4

BLURRED EDGES

Scenario: An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpretation & Cause

Excessive smoothing (e.g., large kernel averaging) reduces high-frequency components, leading to blurred edges and loss of fine details.

Over-Smoothed Image (Input)



Sharpened Image (Output)



(b) Key Issues

- Loss of edge sharpness
- Important details are smoothed out
- Objects boundaries become unclear
- Over-smoothing reduces accuracy

(c) Applied Technique: Unsharp Masking

Formula: $g(x,y) = f(x,y) + k \{ f(x,y) - f_{\text{blur}}(x,y) \}$

f = original image, f_{blur} = blurred image,

k = sharpening factor (e.g., 1.5)



(d) Justification

Unsharp masking enhances edges by boosting high-frequency components while keeping overall image natural.

(e) Observe

Edges are sharper and details are restored without amplifying noise excessively.

Conclusion: Unsharp masking effectively restores edge sharpness lost due to over-smoothing.



5

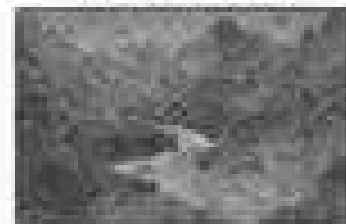
HIGH-FREQUENCY NOISE

Scenario: A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpretation & Description

High-frequency noise appears as rapid intensity fluctuations. It masks fine details and makes interpretation difficult.

Noisy Satellite Image (Input)



(b) Key Issues

- Grainy appearance
- Fine features are obscured
- Poor visual quality
- Affects further processing

Filtered Image (Output)



(c) Applied Technique: Low-Pass Filtering (Gaussian in Frequency Domain)

Idea: Attenuate high-frequency components and preserve low-frequency information.



(d) Justification

Low-pass filter removes high-frequency noise while preserving overall structures and smooth variations.

(e) Filter Used

$$H(u, v) = e^{-\frac{D(u, v)^2}{20^2}}$$

(Gaussian LPF)

Observation: Noise is reduced and features become more visible.

Conclusion: Gaussian LPF is effective for high-frequency noise. ✓

6

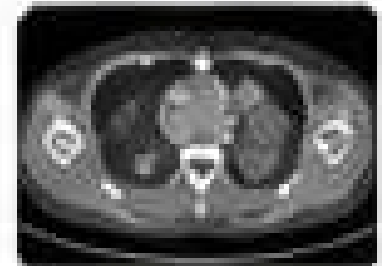
OBJECT BOUNDARY DETECTION

Scenario: In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpretation & Importance

Boundary detection identifies significant intensity changes corresponding to edges. It helps in diagnosis and measurement.

Input (Medical Image)



(b) Key Challenges

- Weak edges due to low contrast
- Noise affects edge detection
- Varying intensity across tissues
- Need for accurate localization

Edges (Output)



(c) Applied Technique: Canny Edge Detection

Steps:



(d) Justification

Canny provides thin, accurate edges with good localization and is robust to noise.

(e) Key Parameters

- Low Threshold
- High Threshold
- Gaussian Sigma

Observation: Boundaries of tissues are clearly identified with minimal noise.



7

SIMPLE SEGMENTATION

Scenario: An image contains objects clearly distinguishable from the background based on intensity.

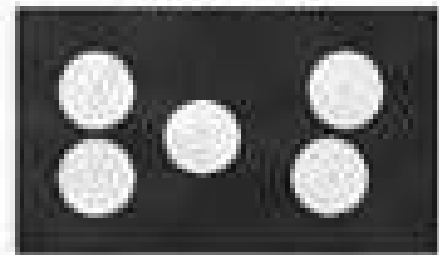
(a) Interpretation & Requirement

Objects and background have distinct intensity ranges.
Segmentation can be performed using thresholding.

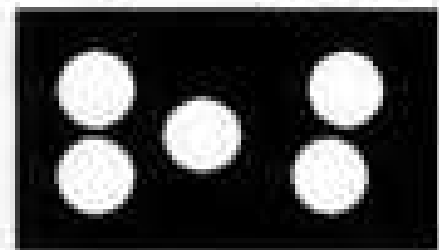
(b) Key Features

- Bimodal histogram
- Clear separation between object and background
- Uniform object intensity

Input Image



Segmented (Output)

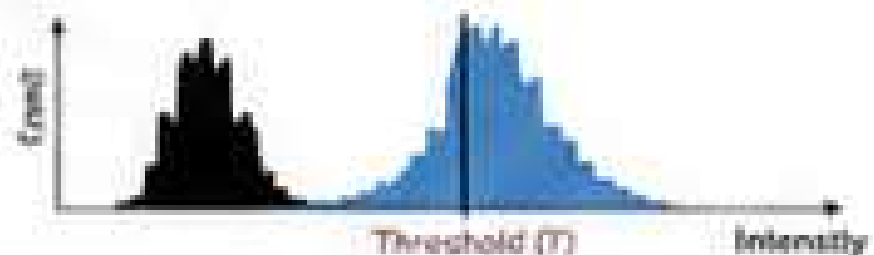


(c) Applied Technique: Global Thresholding

Formula:

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

Histogram



(d) Justification

Global thresholding is simple and effective when the object and background are well separated.

(e) Result

- Objects are successfully
- separated from the
- background.

Observation: Segmented image clearly extracts objects with clean boundaries.



8

COMPLEX SEGMENTATION

Scenario: An image contains multiple regions with overlapping intensity values.

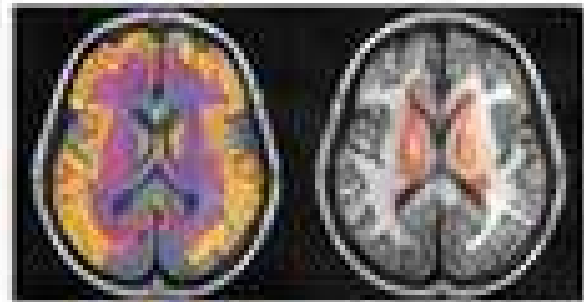
(a) Interpretation & Challenges

Intensity overlap makes simple thresholding ineffective.

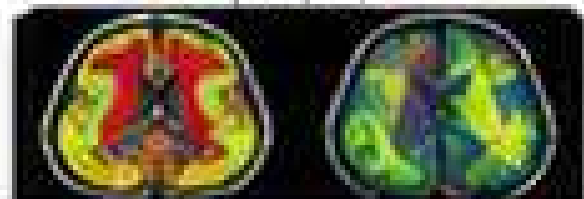
Need advanced methods to separate regions.

(b) Key Issues

- Overlapping intensities
- Multiple regions
- Inhomogeneous areas
- Complex boundaries

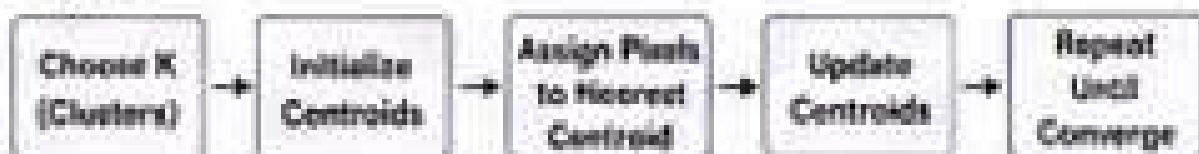


Segmented
(Output)



(c) Applied Technique: K-Means Clustering

Steps:



(d) Justification

K-Means groups pixels based on intensity similarity and effectively handles overlap and multiple regions.

(e) Result

- Different regions are
- segmented into distinct
- clusters.

Observation: Multiple regions are segmented accurately despite intensity overlap.



9

BROKEN EDGES

Scenario: After applying edge detection, object boundaries appear discontinuous.

(a) Interpretation & Cause

Edges are broken due to noise, weak gradients, thresholding, or gaps in edge detection.

(b) Key Issues

- Discontinuous edges
- Gaps in boundaries
- Affects shape analysis
- Reduces accuracy

Edge Map (Broken)



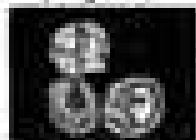
Improved Edges (Output)



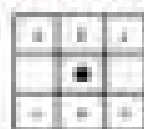
(c) Applied Technique: Morphological Closing

Process: Dilation followed by Erosion (with same structuring element)

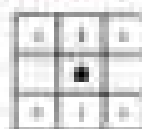
Edge Map (Input)



Dilation



Erosion



Closed Edges (Output)



(d) Justification

Closing connects nearby edge pixels and fills small gaps, improving continuity.

(e) Result

Boundaries become continuous and object shapes are clearer.

Observation: Edge connectivity is improved and gaps are filled.



10

COMBINED REQUIREMENT

Scenario: An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpretation & Requirement

Need to reduce noise but keep edges sharp for accurate defect identification.

(b) Key Issues

- Presence of noise
- Edge preservation necessary
- Over-smoothing hides defects
- Balance between smoothing and detail retention

Noisy Image
(Input)



Enhanced
(Output)



(c) Applied Sequence of Techniques

Recommended Pipeline:



(d) Justification

Median removes noise, bilateral preserves edges during smoothing, and unsharp masking enhances defect visibility.

(e) Final Observation

Noise is reduced and edges are enhanced. Defects are clearly visible.

Conclusion: This sequence achieves noise removal and edge preservation for reliable inspection.

